

ABERDARE TECHNICAL COLLEGE
PRACTICAL OBSERVATION CHECKLIST
NETWORK MODULE 2
PRACTICAL CHECKLIST 2

Candidate name: _____ **Admission no.:** _____

Class/Course: _____ **Date:** _____

Assessor: _____

CUTTING LIST (MATERIALS AND TOOLS LIST)

No.	Item	Specification	Qty per candidate	Provided by
1	Desktop PC or laptop	Windows, working Ethernet port/NIC	3	College
2	Network switch	Minimum 5 ports, unmanaged	1	College
3	Router	Web admin access, reset to factory defaults	1	College
4	Ethernet patch cable	UTP Cat5e/Cat6, RJ45 both ends, about 1.5 m	4	College
5	Power cables/adapters	For router, switch and PCs	As needed	College
6	Router default login sheet	Default IP, username, password	1	Assessor
7	Answer sheet	Printed copy	1	Assessor
8	Target addressing table	Printed copy of IP, mask and gateway values	1	Assessor
9	Pen	Blue or black	1	Candidate

PART A: PRACTICAL OBSERVATION CHECKLIST (50 marks)

Tick **Yes** if the candidate performs the step correctly without help and award the full mark.
Otherwise tick **No** and award 0.

Task 1: Set up the network hardware

No.	Observation item	Max	Yes	No	Score	remarks
1.1	Selects the correct materials and equipment from the cutting list	1				
1.2	Connects the router LAN port to the switch	1				
1.3	Connects PC1, PC2 and PC3 to separate switch ports	1				
1.4	Handles cables and equipment correctly (no damage, cables seated properly)	1				
1.5	Powers on the router, switch and PCs	1				
1.6	Checks link lights on all connected ports	1				
	Total for Task 1	6			<u> </u> / 6	

Task 2: Access the router

No.	Observation item	Max	Yes	No	Score	Remarks
2.1	Opens Command Prompt and runs ipconfig	1				
2.2	Correctly identifies the default gateway	1				
2.3	Enters the gateway address in the web browser	1				
2.4	Logs in to the router using the correct credentials	1				
	Total for Task 2	4			<u> </u> / 4	

Task 3: Change the router's IP address

No.	Observation item	Max	Yes	No	Score	remarks
3.1	Locates and opens the LAN settings	1				
3.2	Changes the router IP to 192.168.50.1	2				
3.3	Sets the subnet mask to 255.255.255.0	1				
3.4	Sets the DHCP range to 192.168.50.100 to 192.168.50.150	1				
3.5	Saves the settings and waits for the router to restart	1				
3.6	Reconnects and opens 192.168.50.1 successfully	1				
	Total for Task 3	7			<u> </u> / 7	

Task 4: Configure IP addresses on the PCs

No.	Observation item	Max	Yes	No	Score	remarks
4.1	Opens the IPv4 properties of the Ethernet adapter	1				
4.2	Configures PC1 correctly (IP, mask, gateway)	2				
4.3	Configures PC2 correctly (IP, mask, gateway)	2				
4.4	Configures PC3 correctly (IP, mask, gateway)	2				
4.5	Sets the DNS server to 8.8.8.8	1				
4.6	Verifies the settings using ipconfig	1				
	Total for Task 4	9			<u> </u> / 9	

Task 5: Test connectivity

No.	Observation item	Max	Yes	No	Score	remarks
5.1	Pings the router from PC1	1				
5.2	Pings PC2 and PC3 from PC1	1				

5.3	Pings PC3 from PC2	1				
5.4	Correctly interprets the ping results (reply vs. timed out)	1				
5.5	Troubleshoots any failure systematically (cables, IP settings, firewall)	1				
	Total for Task 5	5			<u> </u> / 5	

Task 6: Prepare the PCs for file sharing

No.	Observation item	Max	Yes	No	Score	remarks
6.1	Turns on network discovery	1				
6.2	Turns on file and printer sharing	1				
6.3	Sets the network profile to Private	1				
	Total for Task 6	3			<u> </u> / 3	

Task 7: Share a folder on PC1

No.	Observation item	Max	Yes	No	Score	Remarks
7.1	Creates the folder ExamShare	1				
7.2	Creates candidate.txt with name and admission number	1				
7.3	Opens Advanced Sharing and ticks "Share this folder"	1				
7.4	Uses the correct share name ExamShare	1				
7.5	Gives Everyone Read permission	1				
7.6	Applies the settings by clicking OK	1				
	Total for Task 7	6			<u> </u> / 6	

Task 8: Access shared files across the network

No.	Observation item	Max	Yes	No	Score
8.1	From PC2, accesses \\192.168.50.11 correctly	1			
8.2	Opens ExamShare and copies candidate.txt to PC2	1			
8.3	Repeats the access and copy from PC3	1			
8.4	Creates and shares FromPC2 with Read/Write permission on PC2	2			
8.5	From PC3, accesses \\192.168.50.12 and copies hello.txt	1			
	Total for Task 8	6			<u> </u> / 6

General conduct

No.	Observation item	Max	Yes	No	Score
G.1	Works safely and handles equipment responsibly	1			
G.2	Completes the tasks within the allocated time (2 hours)	1			
	Total for General conduct	2			<u> </u> / 2

Total for Part A 50 / 50

PART B: PRODUCT CHECKLIST (10 marks)

Assess the final working network and the candidate's answer sheet.

No.	Product criteria	Max	Yes	No	Score
B.1	Network is physically complete, with neat and secure cabling	1			
B.2	Router admin page opens at the new IP 192.168.50.1	2			
B.3	PC1, PC2 and PC3 have the correct IP, mask and gateway (verified by assessor)	3			
B.4	Ping tests between all devices succeed	1			
B.5	ExamShare is accessible and readable from PC2 and PC3	1			

B.6	From PC2 is accessible from PC3 and the file can be copied	1			
B.7	Answer sheet is fully and accurately completed (cables, ports, default and new IPs, PC settings, ping results)	1			
	Total for Part B	10			<u> </u> / 10

PART C: ORAL CHECKLIST (30 marks)

Ask each question after the practical. Award **3** for a complete correct answer, **2** for a mostly correct answer, **1** for a partial answer, and **0** for no answer or a wrong answer.

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No.	Oral question	Expected answer	Max	Score	remarks
1	What is an IP address?	A logical address that identifies a device on a network, e.g. 192.168.50.11	3		
2	What is the purpose of a subnet mask?	Separates the network part from the host part of an IP address; 255.255.255.0 means the first three numbers identify the network	3		
3	What is a default gateway, and what did you set it to?	The router's address, used to reach other networks and the internet; set to 192.168.50.1	3		
4	Why did you connect the PCs and router to a switch?	A switch connects devices in the same network and forwards data to the correct device; it provides extra ports	3		
5	Why would you change a router's default IP address?	To avoid conflicts with other networks, follow an addressing plan, and improve security	3		
6	What is DHCP, and why did you keep the PC addresses outside the DHCP range?	DHCP assigns IP settings automatically; keeping static addresses outside the range prevents IP conflicts	3		
7	What do ipconfig and ping do?	ipconfig shows IP address, subnet mask and gateway; ping tests connectivity between devices	3		
8	What would you check if a ping fails?	Cables and link lights, IP address/mask/gateway settings, same subnet, firewall blocking ICMP	3		

9	What does \\192.168.50.11 mean, and why did you set Read permission on ExamShare?	A UNC path used to reach a shared folder on another PC; Read lets others view and copy files but not change or delete them	3		
10	Name two security steps for a router or shared network	Any two: change the default admin password, use WPA2/WPA3 with a strong password, update firmware, limit share permissions, keep the firewall on	3		
	Total for Part C		30	<u> </u> / 30	

Assessor's comments:

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COMPETENCY DECISION

Total score	Decision
49 and below	NOT COMPETENT
50 and above	COMPETENT

Final decision (tick one): ☐ COMPETENT ☐ NOT COMPETENT

Assessor's comments:

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Assessor's signature: _____ **Date:** _____

Candidate's signature: _____ **Date:** _____